

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 17 September 2026, 8:30 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:30 AM**Time taken** 50 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  #define MAX_N 100005
4
5  int A[MAX_N];
6
7  int main() {
8      int n;
9      // Read the total number of elements (handles line spaces or newlines automatically)
10     if (scanf("%d", &n) != 1) return 0;
11
12     // Read the n elements of the array
13     for (int idx = 0; idx < n; idx++) {
14         if (scanf("%d", &A[idx]) != 1) return 0;
15     }
16
17     int k;
18     // Read the target non-negative integer difference k
19     if (scanf("%d", &k) != 1) return 0;
20
21     int i = 0;
22     int j = 1;
23     int found = 0;
24
25     // Two-pointer linear traversal
26     while (i < n && j < n) {
27         int diff = A[j] - A[i];
28
29         if (i != j && diff == k) {
30             found = 1;
31             break;
32         } else if (diff < k) {

```

```
33         j++;
34     } else {
35         i++;
36     }
37
38     // Ensure the two pointers never overlap on the exact same index
39     if (i == j) {
40         j++;
41     }
42 }
43
44 // Output 1 if a pair exists, otherwise 0
45 printf("%d\n", found);
46
47 return 0;
48 }
49
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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